MAKSHUDUR RAHMAN

Research Enthusiast | Power Electronics • Control Systems • Metasurfaces • Optical Sensing | Analytical Thinker
Rajshahi University of Engineering & Technology (RUET)
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Bashundhara R/A, Dhaka, Bangladesh

Career Objective

Aspiring to pursue a career in teaching and research with a commitment to exploring emerging ideas and fields, leveraging knowledge to devise practical solutions and making meaningful academic and scientific contributions.

Education

B.Sc. in Electronics & Telecommunication Engineering 2022 – 2026
Rajshahi University of Engineering & Technology (RUET), Rajshahi, Bangladesh
CGPA: **3.73/4.00** (Class position: **5th**)

Higher Secondary Certificate (HSC) 2018 – 2020
Govt. Ananda Mohan College, Mymensingh, Bangladesh
GPA: **5.00/5.00**

Secondary School Certificate (SSC) 2010 –2018
Progressive Model School, Mymensingh, Bangladesh
GPA: **5.00/5.00**

Experience

Bangladesh Telecommunications Company Limited (BTCL) March 2025
Industrial Trainee
Skills: OSP Network, GPON Network, Optical Fiber Splicing & Cable Jointing, OTDR-Based Fiber Loss Measurement, Fiber Fault Detection & Troubleshooting, FTTH Network Operation & Maintenance, Telecommunication Billing System

Publications

Disturbance-Adaptive Nonlinear Damping Control for Transient Stabilization of Grid-Forming Inverters Under Variable Grid Strength. Aug 2026
2026 IEEE International Conference on Power, Electrical, Electronics and Industrial Applications (PEEIACON).

Polarization Insensitive Metasurfaces Absorber for Visible Wavelengths. Dec 2025
28th International Conference on Computer and Information Technology (ICCIT).
DOI:10.1109/ICCIT68739.2025.11491689

Undergraduate Thesis

A Modified Virtual Synchronous Generator Control for Transient Stabilization of Grid-Forming Inverters Under Variable Grid Strength June 2025 – July 2026

Projects

Project 1: Smart Guest House Management System *Jan 2024*
• Developed a full-stack guest house management system with online room booking, secure authentication, Stripe payment integration, and an admin dashboard. GitHub

Project 2: Multifunctional Autonomous Robot Car *Dec 2023*
• Developed an Arduino-based multifunctional robot car with autonomous object-following and obstacle-avoidance capabilities, along with manual control through Wi-Fi, voice commands, and Bluetooth. GitHub

Project 3: Line Following Robot (LFR) *Dec 2022*
• Developed an Arduino-based line-following car. GitHub

Lab Project 1: Pulse Radar System for Multi-Target Detection and Tracking *April 2026*
• Developed a MATLAB/Simulink pulse-radar model for multi-target detection, range estimation, and velocity estimation using range-Doppler analysis. GitHub

Lab Project 2: Advanced Wireless Communication Modeling for 4G/5G and OFDM Systems *Oct 2025*
• Developed MATLAB models for wireless propagation, fading, equalization, multiple access, cellular handoff, and OFDM performance evaluation. GitHub

Lab Project 3: Design and Performance Optimization of a Tin-Based Perovskite Solar Cell *May 2025*
• Developed a SCAPS-1D solar-cell model and analyzed V_{oc}, J_{sc}, FF, PCE, absorber thickness, temperature, doping, and interface recombination for performance optimization. GitHub

Lab Project 4: Simulation and Analysis of Chromatic Dispersion and Birefringence in Optical Fibers *April 2025*
• Simulated chromatic dispersion, attenuation, and birefringence using MATLAB and COMSOL to characterize wavelength-dependent and polarization-dependent behavior. GitHub

Lab Project 5: Design and Performance Optimization of a 2.4 GHz Microstrip Patch Antenna *Jan 2025*
• Designed and simulated a 2.4 GHz microstrip patch antenna in CST, analyzing S_{11}, VSWR, gain, directivity, radiation pattern, and efficiency. GitHub

Lab Project 6: Computational Algorithms for Power-System Network Reduction and Power Flow Analysis *Nov 2024*
• Developed MATLAB algorithms for Y-bus construction, Kron reduction, Gauss-Seidel and Newton-Raphson power flow, and symmetrical-component analysis. GitHub

Lab Project 7: Full-Stack Development of a Dynamic Personal Portfolio Platform *May 2024*
• Developed a responsive portfolio platform using HTML, CSS, JavaScript, PHP, and MySQL for dynamic content and data management. GitHub

Lab Project 8: Development and Experimental Analysis of SCR-Based Power Conversion Systems *Oct 2023*
• Developed and experimentally analyzed SCR-based rectifiers, AC voltage controllers, and cycloconverters using firing-angle and Arduino-based control. GitHub

Skills & Interests

Programming Languages: C, C++, Python, JavaScript, MATLAB, SQL, Assembly 8086, LaTeX.

Simulation & Software: COMSOL Multiphysics, CST Studio Suite, DSCH, Microwind2, EMU-8086, eNSP Simulator, HOMER Pro, SCAPS-1D, Origin Pro, Arduino IDE, AutoCAD, ExpressPCB.

Web Technologies: HTML5, CSS3, Node.js, Express.js, PHP, MySQL.

Productivity Tools: Microsoft Word, Excel, PowerPoint, Google Colab, Visual Studio Code, Code::Blocks, Adobe Illustrator.

Professional Skills: Technical presentation, public speaking, leadership, teamwork, event management, problem-solving, critical thinking, adaptability.

Languages: Bengali (full professional proficiency), English (full professional proficiency), Hindi.

Research Interests: Power electronics; renewable energy; control systems; metasurfaces and nanophotonics; plasmonic absorbers and optical sensing.

Activities & Volunteering

- **RUET Career Forum (RCF)** – General Member June 2022 – July 2024
- **ETECHNOVATION 2022** – Volunteer and Line Following Robot Car Race Participant Dec 2022
- **University Innovation Hub Program – Rajshahi Cohort 6** – Presented a Smart Healthcare and Home Medical Service Management concept April 2024

Workshops & Seminars

- **9th RCF Career Fair 2024** – Career development workshops on CV writing, interview skills, career planning, and industry networking; organized by RUET Career Forum with the Office of Students' Welfare, RUET, and REOSA. April 2024
- **International Conference on Computer & Information Engineering (ICCIE 2022)** – Conference on computer science, information engineering, AI, communication, and emerging technologies. Nov 2022

References

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Declaration

I hereby declare that the information presented in this CV is true. **Makshudur Rahman**